

Marwa AlAlawi

Ph.D. Candidate

MIT Mechanical Engineering Department | MIT Computer Science and Artificial Intelligence Lab
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Education

Massachusetts Institute of Technology, Cambridge, MA 2023 – Present

Ph.D. in Mechanical Engineering
MIT Mechanical Engineering Department
MIT Computer Science and Artificial Intelligence Lab
Advisor: Stefanie Mueller

Massachusetts Institute of Technology, Cambridge, MA 2021 – 2023

M.Sc. in Mechanical Engineering
MIT Mechanical Engineering Department
MIT Computer Science and Artificial Intelligence Lab
Advisor: Stefanie Mueller

Massachusetts Institute of Technology, Cambridge, MA 2016 – 2020

B.Sc. in Mechanical Engineering & Concentration in Japanese Studies
MIT Mechanical Engineering Department
Additional Coursework in EECS
Advisor: Brian Anthony

Publications

*In Human Computer Interaction (HCI), **peer-reviewed conference papers** are the **primary, most prestigious** venues. The premier venues are **ACM CHI** (h5-index: 139) and **ACM UIST** (leading technical HCI venue, h5-index: 53)*

Full Paper Publications (top tier venues in HCI)

- [5] **Marwa AlAlawi**, Regina Zheng, Abdullah Negm, Jiaji Li, Emma Li, Yasuaki Kakehi, Yoshihiro Kawahara, Junyi Zhu, Ticha Sethapakdi & Stefanie Mueller. “Bifur-circuits: Interactive and Modular Metamaterial Building Blocks Via Bifurcated Geometries”, *In Proceedings of ACM Symposium on User Interface Software and Technology (UIST) 2026 (to appear)*.
- [4] **Marwa AlAlawi**, Regina Zheng, Sooyeon Ahn, Katherine Yan, Ticha Sethapakdi, Junyi Zhu & Stefanie Mueller. “Meta-antenna: Mechanically Frequency Reconfigurable Metamaterial Antennas,” *In Proceedings of the ACM Symposium on User Interface Software and Technology (UIST) 2025*, Article 36, 1–19. | [Project Page](#)
- [3] Jiaji Li, Tianshu Dong, Mingming Li, Zhiqi Wang, Xin Tang, **Marwa AlAlawi** & Stefanie Mueller. “XpendFab: 3D Printing Expendable Mechanisms for Functional Breakage, Splitting, and Disassembly”, *Design and Artificial Intelligence 2025*, vol. 1, no. 4, Article 100051.

- [2]  Mehmet Ozdemir, **Marwa AlAlawi**, Mustafa Doga Dogan, Jose F. Martinez Castro, Stefanie Mueller & Zjenja Doubrovski. “Speed-Modulated Ironing: A Method for High-Resolution Shade and Texture Gradients in Single-Material 3D Printing,” *In Proceedings of ACM Symposium on User Interface Software and Technology (UIST) 2024*, Article 121, 1 - 13. **[BEST DEMO NOMINEE]** | [Project Page](#)
- [1] **Marwa AlAlawi**, Noah Pacik-Nelson, Junyi Zhu, Ben Greenspan, Andrew Doan, Brandon Wong, Benjamin Owen-Block, Shanti Mickens, Wilhelm Schoeman, Michael Wessely, Andreea Danielescu & Stefanie Mueller. “MechSense: A Design and Fabrication Pipeline for Integrating Rotary Encoders into 3D Printed Mechanisms,” *In Proceedings of ACM Conference on Human Factors in Computing Systems (CHI) 2023*, Article 626, 1 - 14. | [Project Page](#)

Extended Abstracts (posters & demos at top tier venues in HCI)

- [6] **Marwa AlAlawi**, Faraz Faruqi, Stefanie Mueller & Ticha Sethapakdi. “SensorGen: Prompt-Driven Exploratory Design and Synthesis for Floating Capacitor Sensor Geometries”. *In Proceedings of ACM Symposium on User Interface Software and Technology (UIST) Extended Abstracts 2026 (to appear)*.
- [5]  **Marwa AlAlawi**, Kexin Wang, Regina Zheng, Adelene Chan, Martin Feick & Stefanie Mueller. “Ori-TENG: 3D Printed Origami Tessellations as Triboelectric Nanogenerators for Self-powered Sensing and Energy Harvesting” *In Proceedings of ACM Symposium on User Interface Software and Technology (UIST) Extended Abstracts 2025*, Article 153, 1 - 3. **[BEST POSTER NOMINEE]** | [Project Page](#)
- [4] **Marwa AlAlawi**. “Hybrid Functional Identities for Mechanical Elements”. *In Proceedings of ACM Conference on Human Factors in Computing Systems (CHI) Extended Abstracts, Doctoral Consortium 2025*, Article 829, 1 - 5.
- [3] Doga Ozbek, **Marwa AlAlawi** & Michael Wessely. “AcceloPrint: Fabricating Customizable Accelerometers with Multi-Material 3D Printing,” *In Proceedings of ACM Conference on Human Factors in Computing Systems (CHI) Extended Abstracts 2025*, Article 59, 1 - 8.
- [2] Doga Ozbek, **Marwa AlAlawi**, Till Max Eckroth & Michael Wessely. “Demonstration of AcceloPrint: Fabricating Accelerometers with Multi-Material 3D Printing,” *In Proceedings of the ACM Symposium on Computational Fabrication (SCF) Extended Abstracts 2024*, Article 8, 1 - 3.
- [1] **Marwa AlAlawi** & Brandon Wong. “LeviCircuits: Adhoc Electrical Circuit Prototyping using Ultrasound Levitation,” *In Proceedings of ACM Symposium on User Interface Software and Technology (UIST) Extended Abstracts, Student Innovation Contest 2022*, Article 103, 1 - 3.

Workshops (as organizer)

- [1] **Marwa AlAlawi**, Ticha Sethapakdi, Dishita Turakhia, Katherine Song, Ilan Moyer, Liang He, Lvmin Zhang, Ruofei Du & Stefanie Mueller. “The AI-Augmented Maker: Challenges and Opportunities in Designing Intelligence for Community-driven DIY Cultures” *ACM Symposium on User Interface Software and Technology (UIST) Workshops 2026 (to appear)*.

Selected Press Features

- [9] **Arduino**, Morphing Meta-antennas Enable Frequency Manipulation

2025

- [8] **MIT News**, A Shape-changing Antenna for More Versatile Sensing and Communication
[SPOTLIGHT PIECE] 2025
- [7] **Interesting Engineering**, ‘Bunny ears’: MIT creates shape-changing antenna to transform wireless tech 2025
- [6] **MIT News**, New 3D Printing Technique Creates Unique Objects Quickly and with Less Waste
[SPOTLIGHT PIECE] 2024
- [5] **TU Delft News**, New 3D Printing Technique by TU Delft and MIT Delivers High-Resolution Textures from a Single Material 2024
- [4] **All About Circuits**, MIT Creates New Way to Incorporate Sensors Into 3D-printed Designs 2023
- [3] **3D Printing.com**, MIT Develops 3D Printed Sensors that Sense Rotation 2023
- [2] **MIT News**, 3D-Printed Revolving Devices can Sense How They are Moving 2023
- [1] **Hackster.io**, MIT Engineers Use Ultrasound Levitation to Prototype Circuits 2022

Research Internships

University of Tokyo, Japan

Research Advisors: Yasuaki Takechi & Yoshihiro Kawahara 2025

MIT Mechanical Engineering (Device Realization Group), USA

Research Advisor: Brian Anthony 2019 – 2020

Personal Robots Group (MIT Media Lab), USA

Research Advisor: Cynthia Breazeal 2019

MIT Mechanical Engineering (Biomimetics Group), USA

Research Advisor: Sangbae Kim 2017

Work Experience

REACT Neuro, Cambridge, MA, USA

Virtual Reality Experience Designer

Supervisor: Shaun Patel 2020 – 2021

GREE, Tokyo, Japan

XR Intern

Supervisor: Ken Watanabe 2018

Professional Service

Associate Chair

- **ACM TEI** – ACM Conference on Tangible, Embedded, and Embodied Interaction 2027

Academic Reviewer

- **ACM CHI** – ACM Conference on Human Factors in Computing Systems 2025–2026
- **ACM SIGGRAPH** – SIGGRAPH Posters 2024
- **ACM UIST** – ACM Symposium on User Interface Software and Technology 2024–2026
- **HCI Journal** – Human-Computer Interaction Journal 2023, 2024
- **ACM DIS** – ACM Conference on Designing Interactive Systems 2023, 2025
- **Elsevier C&G** – Computers & Graphics Journal 2026
- **ACM TEI** – ACM Conference on Tangible, Embedded, and Embodied Interaction 2025–2026
- **ACM SCF** – ACM Symposium on Computational Fabrication 2022, 2024, 2026

*Received 5 special recognitions for Outstanding Reviews at top-tier venues (ACM CHI '26; ACM UIST '24, '25).

Chairing

- **ACM SCF** – Student Volunteer Co-chair, ACM Symposium on Computational Fabrication 2026

Student Volunteer

- **ACM UIST** – ACM Symposium on User Interface Software and Technology 2022

Teaching Experience

Co-instructing / Curriculum Development

- [6] **Co-Founder of MIT-Bahrain Global Teaching Lab (GTL)**, MIT 2023 – Present
 - Established the first MIT STEAM initiative in the Gulf region; designed open-source university-level HCI curricula spanning robotics, wearable technology, and digital fabrication, delivering hands-on instruction to 200+ Bahraini high school and undergraduate students across multiple cohorts.
- [5] **Actuated Xperiences**, MIT Bahrain Global Teaching Lab Winter 2025
Lead Instructor & Course Developer | *Course Website:* <https://mit-gtl-bahrain-2025.github.io/>
 - Managed end-to-end execution of an intensive 1-month accelerated engineering design course (10 core lectures, mini-projects, and final deliverables) for 55 university students and a 5-person teaching team; handled curriculum design (3D modeling, sensing, robotics etc), scheduling, logistics, lecturing, and technical mentorship of 9 student team projects (e.g., underwater drone, social robot); class sponsored by Bank of Bahrain & Kuwait and hosted at American University of Bahrain (AUBH).
- [4] **Introduction to Design Through Engineering Process**, MIT Japan Summer 2024–2026
Lead Instructor
 - Taught structural engineering design challenges *in Japanese* to 100+ middle school students; in partnership with MIT Japan, the City of Kesennuma, and Japanese NPOs.
- [3] **Introduction to Wearable Technology**, MIT Bahrain Global Teaching Lab Winter 2024
Course Director
 - Directed and supervised overall course strategy, operations, hardware procurement, and teaching team selection (with MISTI) for 50 university students; program funded by the Bank of Bahrain and Kuwait (BBK).

[2] **Robotics from Scratch** – MIT Bahrain Global Teaching Lab Winter 2023
Course Co-Director

- Co-secured inaugural program grants from the U.S. Embassy in Bahrain, Riffa Views International School, and AUBH, and co-directed end-to-end curriculum design, procurement, and instructor recruitment for 40 high school and university students.

[1] **Introduction to Making**, MIT Jordan Global Teaching Labs Winter 2020
Co-Instructor & Curriculum Developer

- Co-developed and taught a month-long curriculum for 60+ Jordanian high school students covering 3D modeling/printing, VR development, programming, and circuit design for product development.

Teaching Assistant

[6] **MIT 2.810 Manufacturing Processes and Systems** (MechE, Graduate) Fall 2026
Taught by Josh Ramos and John Lewandowski

[5] **MIT 2.00 Introduction to Design** (MechE, Undergraduate) Fall 2024
Taught by Maria Yang

[4] **MIT 2.00b Toy Product Design** (MechE, Undergraduate) Spring 2022
Taught by David Wallace

[3] **MIT 2.009 Product Engineering Processes** (MechE, Undergraduate Capstone) Fall 2020
Taught by David Wallace

[2] **MIT 2.00b Toy Product Design** (MechE, Undergraduate) Spring 2020
Taught by David Wallace

[1] **MIT 8.01 Classical Mechanics** (Physics, Undergraduate) Fall 2017
Taught by Peter Dourmashkin

Manufacturing Mentorship

MIT Makerworkshop — Student Machine Shop Mentor 2019 – 2022

MIT Makerlodge — Student Machine Shop Mentor 2017 – 2018

Mentoring

All MIT students are co-advised by Professor Stefanie Mueller.

Masters Students (MEng)

[1] Abdullah Negm 2025 – 2026

Undergraduate Research Assistants (UROP)

[1] Wilhelm Schoeman	2022	[7] Katherine Yan	2024
[2] Shanti Mickens	2022	[8] Audrey Garon	2024
[3] Benjamin Owen-Block	2022	[9] Nathan Syvash	2024
[4] Andrew Doan	2022	[10] 🏆 Regina Zheng*	2024 – 2026
[5] Brandon Wong	2022 – 2023	[11] Emma Li	2025 – 2026
[6] Adelene Chan	2023		

*MIT MechE Carl G. Sontheimer Prize for Best Undergraduate Researcher

Awards & Recognitions

Fellowships & Honors

[4] **Crown Prince International Scholarship, Bahrain** 2014 – Present
Most prestigious national scholarship (10 awarded/yr); funds undergraduate through PhD.

[3] **Wunsch & Silent Hoist Award, MIT Department of Mechanical Engineering** 2026
Awarded for outstanding graduate student excellence across all MIT Mechanical Engineering disciplines.

[2] **JST ASPIRE Fund Recipient, Japan Science and Technology Agency** 2025
International research award funding summer research internship at the University of Tokyo.

[1] **Pi Tau Sigma (PTS) Mechanical Engineering Honor Society, MIT**
National Mechanical Engineering Honor Society.

Research Awards

[3] **ACM UIST Best Poster Nominee (Jury Choice), Korea** 2025
Nominated for best poster presentation at ACM UIST.

[2] **ACM CHI 2025 Doctoral Consortium, Japan** 2025
Highly selective, peer-reviewed venue for 21 PhD students in HCI to present research to leading field experts.

[1] **ACM UIST Best Demo Nominee (People's Choice), USA** 2024
Nominated for best live demonstration at ACM UIST.

Service, Arts & Leadership Awards

[4] **MIT MISTI Ambassador Award, MIT** 2023
Awarded to 0–1 student annually; nominated by the MIT Japan and MIT Arab World programs for leading international collaboration initiatives.

[3] **MIT Mind Hand Heart Grant Recipient, MIT** 2024
Awarded grant for “*Research, Reflections and Rants: A Comic Book About MIT Graduate Student Experiences*”.

[2] **MIT Arts Grant Recipient, MIT** 2024
Awarded grant to build month-long exhibition (*Interwoven*) on sustainability at the MIT Wiesner Gallery and MIT Museum.

[1] **1st Place, 6th Annual New England Korean Speech Competition, MA** 2025
Hosted by the Consulate General of the Republic of Korea in Boston; Represented MIT Global Languages.

Invited Talks

Research & Technical Talks

- [6] **University of Tokyo**, hosted by Takeo Igarashi and Koya Narumi
Digital Fabrication 2025
- [5] **University of Tokyo**, hosted by Takeo Igarashi and Koya Narumi
A Design & Fabrication Pipeline for 3D Printing Rotary Encoders 2024
- [4] **3D Micro Printing**, hosted by Mike Pinelis
3D-Printed Revolving Devices with Integrated Sensors 2023
- [3] **American University of Bahrain**, hosted by Bradley J. Cook
Novel Material Technologies 2023, 2024
- [2] **MIT Nano**, hosted by Brian Anthony
Design of a Virtual Environment for Physiological and Subjective Monitoring of User Presence in VR 2020
- [1] **University of Bahrain**, coordinated by Shatha Ahmad
Using Optical Character Recognition to Build a Physical Japanese Text Translator 2019

Global Leadership, Outreach & Panel Presentations

- [9] **MIT Center for International Studies**, hosted by Chris Pilcavage (MIT Japan)
Beyond 3:11: Tohoku Moving Forward 2026
- [8] **Korean Consulate of Boston**, ft. National Peace Corps
Friends of Korea Event (guest speaker) 2026
- [7] **Bahrain National Television**, hosted by Bahrain Now Variety Show
MIT Global Teaching Labs in Bahrain 2025
- [6] **MIT MISTI**, hosted by MIT Japan and MIT Arab World
MISTI Tech Re-unions: Supporting Global Changemakers 2023
- [5] **MIT**, hosted by MIT Japan & MIT South Asia
International Women's Day Panel: Networking & Building Community 2023
- [4] **MIT**, hosted by Eduardo Rivera
MISTI Alumni Career Panel: Career, Culture and Language Barriers 2021
- [3] **CleverPlay**, hosted by Latifa AlKhalifa
The STEMCast: Mechanical Engineering Through HER Eyes 2021
- [2] **National Space Science Agency**
NSSA Symposium: Women in STEM – Investigating Space Through Virtual Reality (keynote speaker) 2021

[1] **The Hill Editorial**, hosted by Steve Clemons & Embassy of the UAE
Aspiration and Resilience: The New Space Age

2021

Other Academic & Community Service

President, MIT Mechanical Engineering Graduate Association of Women (MEGAWomen) 2025 – 2026

Bilingual Guest Liaison & Translator, Boston Cultural Exchange Initiatives 2023, 2024, 2026
Provided Japanese-English consecutive interpretation and liaison management for visiting Japanese delegates

Cultural Program Coordinator, Showa Japan Women's University Friendship Circle, Boston, MA
2024

International Mentor, Showa Japan Women's University College Connection, Boston, MA 2022 – 2023

Student Mentor, Japanese High School Student Exchange Fellowship, Boston, MA 2023

Selected Service and Teaching Press

[5] **MIT News**, Scene at MIT: Artfinity Brings Artistic Celebration to Campus 2025

[4] **Arts at MIT**, Craft, Community, and Climate 2024

[3] **MIT Science & Technology Initiative**, MISTI, A Launchpad Into MIT Students' Careers
While Helping Them Become Global Citizens 2023

[2] **Bahrain News Agency**, In a First for GCC, MISTI Program to Be Launched in Bahrain 2022

[1] **MIT Japan Program**, Marwa AlAlawi, MechE, Class of 2020 @ Gree Japan 2022

Languages

- Arabic (Native Fluent)
- English (Bilingual Fluent)
- Japanese (Advanced, JLPT N2 Certification)
- Chinese (Intermediate, 3+ years, Ongoing)
- Korean (Intermediate, 3+ years, Ongoing)

References

Prof. Stefanie Mueller (PhD Advisor)
Associate Professor, EECS & MechE
MIT
stefanie.mueller@mit.edu

Prof. Maria C. Yang
Professor & Deputy Dept. Head, MechE
MIT
mcyang@mit.edu

Prof. David R. Wallace

Professor, MechE

MIT

drwallac@mit.edu

Prof. Scott E. Hudson

Professor, HCII

Carnegie Mellon University

scott.hudson@cs.cmu.edu

Prof. Yasuaki Kakehi

Professor, III

The University of Tokyo

kakehi@iii.u-tokyo.ac.jp